TASK 2: Robotic Arm Model and Pick-and-Place Simulation

1. Objective

To design a 3-D model of a robotic arm, define its degrees of freedom (DOF), and simulate a basic pick-and-place motion. The arm was modelled and simulated in Python (NumPy and Matplotlib) using forward and inverse kinematics; the complete script (arm_sim.py) is submitted with this report. The same dimensions given below can be used to build the arm in SolidWorks or AutoCAD.

2. Degrees of Freedom

Degrees of freedom are the number of independent motions a mechanism can perform. Each revolute joint adds one DOF. The designed arm is an articulated (RRRR) manipulator with **4 DOF** in the arm and one additional gripper action (open/close), which is not counted as a positioning DOF.

Joint	Type	Motion	Purpose
J1	Revolute	Base rotation (yaw) about vertical Z	Swings the arm toward the pick or place point
J2	Revolute	Shoulder pitch	Raises / lowers the upper arm (L2)
J3	Revolute	Elbow pitch	Extends / folds the forearm (L3)
J4	Revolute	Wrist pitch	Keeps the gripper pointing vertically down
G	Prismatic (parallel jaw)	Finger open / close	Grips and releases the object

Table 2: Joints and degrees of freedom

3. Design Dimensions

Parameter	Value	Note
Base / shoulder height (d1)	200 mm	Column on the base plate
Upper arm length (L2)	300 mm	Aluminium link
Forearm length (L3)	250 mm	Aluminium link
Wrist-to-tip length (L4)	100 mm	Wrist plus gripper
Maximum horizontal reach (L2+L3)	550 mm	Fully stretched
Object	40 mm cube	Pick at (300, −200) mm; place at (100, 300) mm

Table 3: Arm dimensions

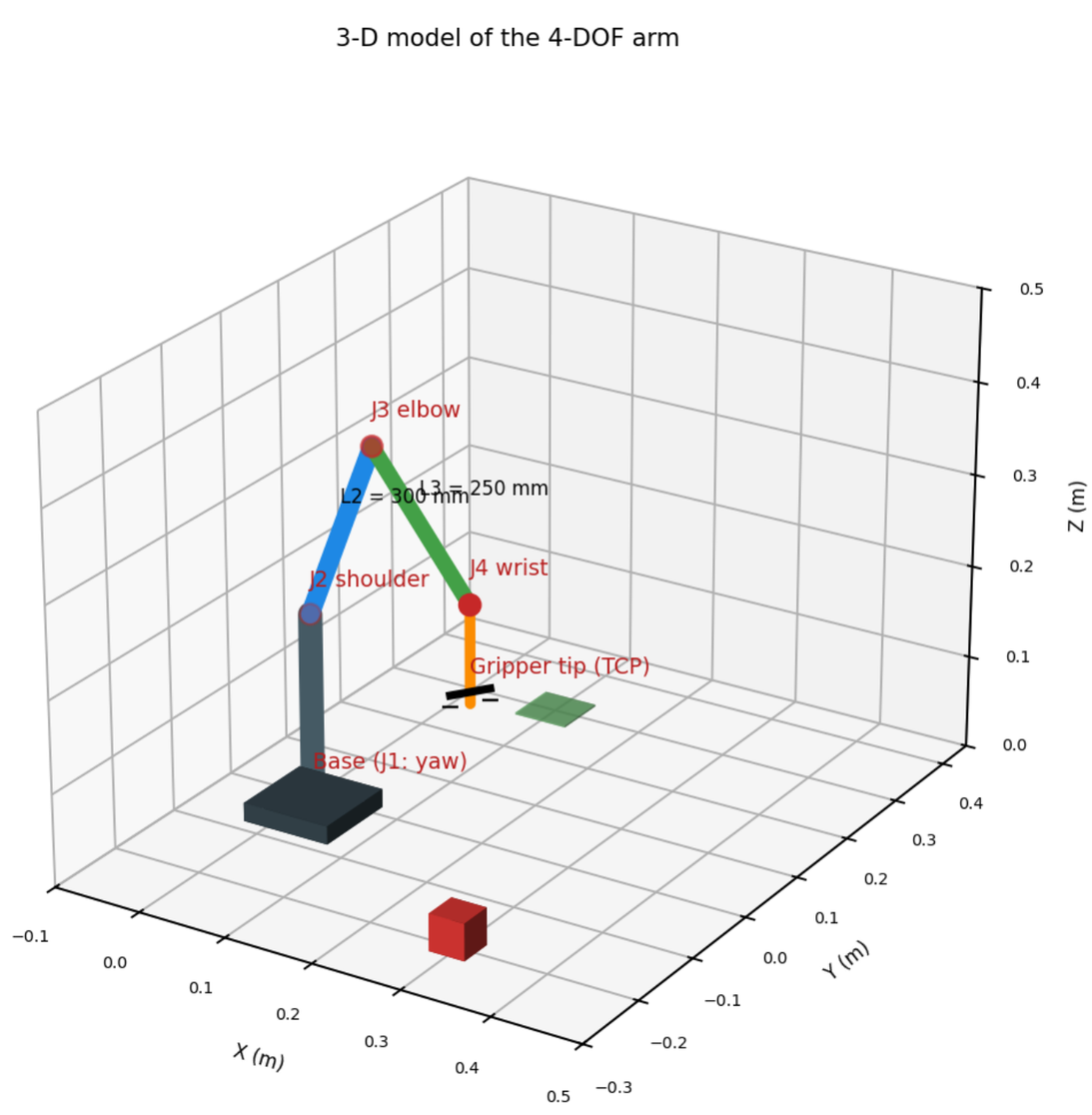


Figure 1: 3-D model of the 4-DOF robotic arm with joints and link lengths

4. Kinematics

Forward kinematics. With joint angles q1 (base), a2 (shoulder), a3 (elbow) and a4 (wrist), the horizontal reach r and height z of the tool tip are:

$$r = L2\cdot\cos(a2) + L3\cdot\cos(a2+a3) + L4\cdot\cos(a2+a3+a4)$$

$$z = d1 + L2\cdot\sin(a2) + L3\cdot\sin(a2+a3) + L4\cdot\sin(a2+a3+a4)$$

$$x = r\cdot\cos(q1), \quad y = r\cdot\sin(q1)$$

Inverse kinematics. For a target (x, y, z) with the gripper vertical ($a2+a3+a4 = -90°$), the base angle is $q1 = \text{atan2}(y, x)$, and the wrist point is at $r = \sqrt{(x^2+y^2)}$, $z' = z + L4 - d1$. The elbow angle follows from the law of cosines, $a3 = -\arccos[(r^2 + z'^2 - L2^2 - L3^2) / (2\cdot L2\cdot L3)]$ (elbow-up solution), and the shoulder angle is $a2 = \text{atan2}(z', r) - \text{atan2}(L3\cdot\sin a3, L2 + L3\cdot\cos a3)$. Finally $a4 = -90° - (a2 + a3)$.

5. Pick-and-Place Motion Sequence

The motion is programmed as a list of waypoints. For each waypoint the inverse kinematics gives the joint angles, and the controller moves between waypoints in joint space. Approaching from above (safe height 250 mm) before descending prevents the gripper from knocking the object or the table.

Waypoint	q1 (°)	a2 (°)	a3 (°)	a4 (°)	Tip (x, y, z) mm	Gripper
Home	0.0	70.0	-110.0	-50.0	(294, 0, 221)	Open
Above pick	-33.7	62.4	-90.0	-62.4	(300, -200, 250)	Open
Grasp (descend & close)	-33.7	29.8	-96.2	-23.6	(300, -200, 20)	Closed
Lift	-33.7	62.4	-90.0	-62.4	(300, -200, 250)	Closed
Above place	71.6	69.8	-101.5	-58.3	(100, 300, 250)	Closed
Place (descend & open)	71.6	32.6	-107.9	-14.7	(100, 300, 20)	Open
Retreat	71.6	69.8	-101.5	-58.3	(100, 300, 250)	Open
Return home	0.0	70.0	-110.0	-50.0	(294, 0, 221)	Open

Table 4: Joint angles at each waypoint (computed by the simulation)

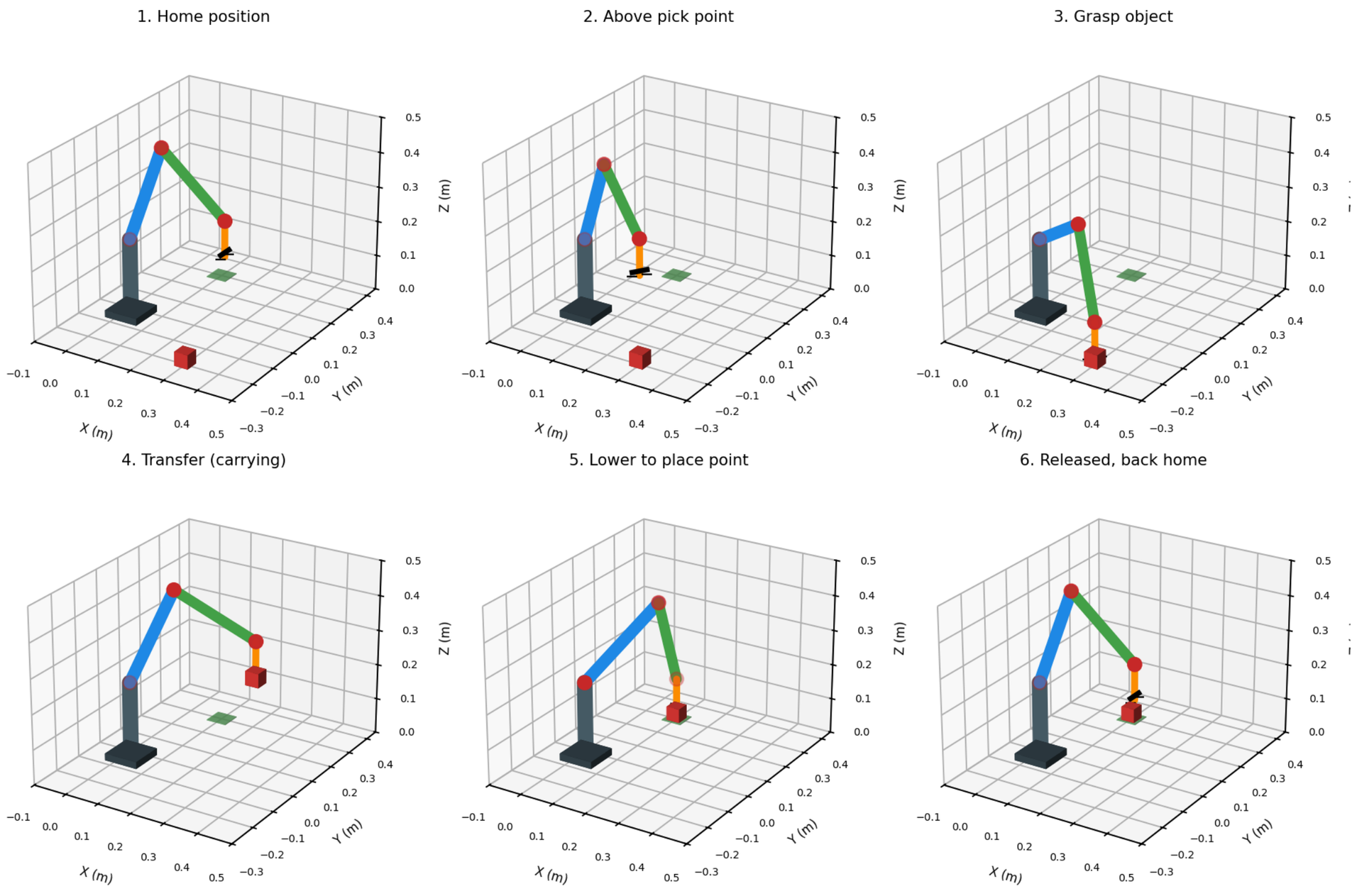


Figure 2: Simulation screenshots of the pick-and-place cycle (red cube = object, green pad = target)

Sequence: (1) start at home; (2) rotate base and move above the object; (3) descend and close the gripper; (4) lift to the safe height; (5) swing to the target with the object; (6) descend, open the gripper to release; (7) retreat upward; (8) return home. All waypoints are inside the 550 mm reach, so the inverse kinematics has a valid solution at every step.

6. Result and How to Reproduce in CAD

The simulation shows the arm picking the cube from (300, −200) mm and placing it on the target pad at (100, 300) mm, with the gripper staying vertical throughout. To model it in SolidWorks, create the base column, two links and wrist as separate parts with the lengths in Table 3, assemble them with revolute mates at J1–J4, and use the joint angles of Table 4 as the motion study keyframes; the screenshots in Figure 2 can then be replaced or supplemented with SolidWorks captures if the submission specifically requires them.